Cthulhu Node Setup Guide

Connecting your local core wallets as blockchain data providers for Cthulhu

Overview

This guide walks you through exposing your local Mazacoin and Dogecoin core wallets so the Cthulhu backend can use them as a **primary data source**, falling back to public blockchain explorers only when your node is offline.

Coin	Default RPC Port	Config File Location (Windows)	Config File Location (Linux/Mac)
Mazacoin	12832	`%AppData%\Mazacoin\mazacoin.conf`	`~/.mazacoin/mazacoin.conf`
Dogecoin	22555	`%AppData%\Dogecoin\dogecoin.conf`	`~/.dogecoin/dogecoin.conf`

> **Note:** These are the same wallets SUP communicates with, so most of the RPC infrastructure should already be in place. You may only need to verify or adjust a few settings.

1 Verify RPC Configuration

Open each wallet's config file and ensure these lines are present. Create the file if it doesn't exist.

Mazacoin (`mazacoin.conf`)

```
server=1
rpcuser=cthulhu
rpcpassword=YOUR_STRONG_PASSWORD_HERE
rpcport=12832
rpcallowip=127.0.0.1
txindex=1
```

Dogecoin (`dogecoin.conf`)

```
server=1
rpcuser=cthulhu
rpcpassword=YOUR_STRONG_PASSWORD_HERE
rpcport=22555
rpcallowip=127.0.0.1
txindex=1
```

> **Important:** If `txindex=1` was not previously set, the wallet will need to **reindex the entire blockchain** on next startup. Launch with the `-reindex` flag once:
>
> `mazacoind -reindex` or `dogecoind -reindex`
>
> This can take several hours depending on your hardware. Only needed once.

Restart both wallets after saving the config changes.

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2 Test RPC Locally

Before exposing anything to the internet, verify RPC works on your machine. Open a terminal/command prompt:

Test Mazacoin

```
curl -u cthulhu:YOUR_PASSWORD -X POST http://localhost:12832 \
  -H "Content-Type: application/json" \
  -d "{\"method\":\"getblockcount\",\"params\":[],\"id\":1}"
```

Test Dogecoin

```
curl -u cthulhu:YOUR_PASSWORD -X POST http://localhost:22555 \
  -H "Content-Type: application/json" \
  -d "{\"method\":\"getblockcount\",\"params\":[],\"id\":1}"
```

> **Expected response:** A JSON object containing `"result": 123456` (a block height number). If you see this, RPC is working.

If you get a connection error, the wallet may not be running or the config wasn't loaded. Double-check the conf file path and restart.

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3 Expose RPC to the Internet

The Cthulhu server needs to reach your wallet's RPC. Below are two options, ordered by recommendation.

Option A: Cloudflare Tunnel (Recommended â€” Free, No Open Ports)

Cloudflare Tunnel creates a secure outbound connection from your machine. Nothing is exposed on your router.

1. **Install** `cloudflared`
   Download from: developers.cloudflare.com/cloudflare-one/connections/connect-networks/downloads
   Windows: Download the .msi installer and run it.
2. **Start a tunnel for Mazacoin**

```
cloudflared tunnel --url http://localhost:12832
```

This prints a public URL like: `https://random-words-here.trycloudflare.com`
Copy this URL.

3. **Start a tunnel for Dogecoin** (in a separate terminal)

```
cloudflared tunnel --url http://localhost:22555
```

Copy this URL too.

> **Note:** These "quick tunnel" URLs change each time you restart `cloudflared`. For a permanent URL, create a named tunnel with a Cloudflare account (free). But for testing, the quick tunnel is perfect.

Option B: ngrok (Alternative â€” Free Tier Available)

1. **Install ngrok**
   Download from: ngrok.com/download
   Sign up for a free account and run `ngrok config add-authtoken YOUR_TOKEN`
2. **Start tunnels**

```
ngrok http 12832    # Mazacoin
ngrok http 22555    # Dogecoin (separate terminal)
```

Copy the `https://xxxx.ngrok-free.app` URLs.

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4 Send Me the Details

Once your tunnels are running, share the following with me:

Item	Example
Mazacoin tunnel URL	`https://abc-xyz.trycloudflare.com`
Dogecoin tunnel URL	`https://def-uvw.trycloudflare.com`
RPC username	`cthulhu`
RPC password	`(your chosen password)`

I will then:

1. **Test connectivity** from the Cthulhu server to your nodes
2. **Verify RPC methods** we need are available (`getrawtransaction` , `listunspent` , `sendrawtransaction` , `getblockcount`)
3. **Discuss next steps** for wiring it into Cthulhu as a priority fallback

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Quick Reference: RPC Methods Cthulhu Needs

Method	Purpose
`getblockcount`	Health check â€" verify node is synced
`getrawtransaction`	Fetch full transaction data (requires `txindex=1`)
`listunspent`	Get UTXOs for an address
`sendrawtransaction`	Broadcast signed transactions to the network
`getblock`	Get block data for confirmation checks
`decoderawtransaction`	Debug/inspect raw transaction hex

Security Reminder: Your RPC credentials grant full wallet access. Never share them publicly. The Cloudflare Tunnel approach is preferred because it doesn't require opening ports on your router. When we integrate this into Cthulhu, the credentials will be stored server-side in environment variables, never exposed to the frontend.

That's it! No rush â€” just follow these steps whenever you're ready to test, and ping me with the tunnel URLs. We'll verify connectivity together and then plan the integration.